Technical Note: IVCP Route Optimizer

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| 1. Abstract

The **Iterative Value-Centric Perturbation (IVCP)** algorithm is a high-performance routing engine designed to provide near-optimal solutions to the **Traveling Salesperson Problem (TSP)** [1] under priority-based operational constraints. Rather than seeking a singular global optimum through exhaustive computation, IVCP utilizes node values to establish a prioritized origin before applying a deterministic **2-Opt** [2] refinement process. This approach is specifically engineered to resolve geometric path inefficiencies in real-time, delivering commercially significant reductions in Euclidean distance for dynamic logistics environments.

Keywords: Graph Theory, Vehicle Routing Problem, Iterative Value-Centric Perturbation, Traveling Salesperson Problem, 2-Opt, 2D Euclidean Coordinate System, Hamiltonian Cycle.

| 2. Problem Definition

The system operates on a set of N cities within a **2D Euclidean Coordinate System** [3]. Each city C_i is defined by a tuple (x_i, y_i, V_i), where V_i represents a priority value between 1 and 10.

- **Primary Objective**: Minimize the total round-trip distance $L = \sum d(C_i, C_{i+1 \; pmod \; N})$.
- **Constraint**: The route must initiate at the node with the highest priority value ($argmax$(V)).

| 3. Algorithm Architecture

The IVCP engineering follows a two-phase execution model:

Phase 1: Value-Driven Initialization

The algorithm identifies the city with the highest value as the starting anchor. It then constructs an initial **Hamiltonian Cycle** [4] using a value-driven nearest neighbor approach: from the current city, it moves to the closest unvisited city based on Euclidean distance.

Phase 2: Deterministic 2-Opt Refinement

The system enters a greedy loop that performs segment reversals. It examines every possible pair of non-adjacent edges (i, $i+1$) and (j, $j+1$). If untangling these edges by reversing the segment between them results in a shorter total path, the change is accepted. This continues until a local optimum is reached and no further improvements are possible.

| 4. Mathematical Foundation

- **Distance Metric**: $d(C_1, C_2) = \sqrt{((x_1 - x_2)^2 + (y_1 - y_2)^2)}$
- **Refinement Condition**: A segment reversal between indices i and j is committed if:
- $d(C_i, C_j) + d(C_{i+1}, C_{j+1}) < d(C_i, C_{i+1}) + d(C_j, C_{j+1})$.

| 5. Conclusion

To validate the robustness of the IVCP algorithm, the performance was measured using a batch-simulation testbench (`IVCP_Route_Optimizer_Testbench.html`). Unlike traditional static tests, this environment subjects the heuristic to **100 consecutive iterations** with high environmental variance:

- **Variable Node Density:** Each iteration randomly scales between **5 and 50 nodes**, simulating everything from small local courier loops to complex regional distribution routes.
- **Scale Variance:** The coordinate plane is dynamic, forcing the algorithm to resolve geometric inefficiencies in both high-density clusters and sparse distributions.
- **Cumulative Accuracy:** Efficiency is measured as a **Net Weighted Gain**—the ratio of total distance saved across all 100 maps versus the total initial greedy distance.

Optimization Result
`Net Efficiency Gain:` **`13.26%`**

These results demonstrate that IVCP successfully recaptures significant operational waste often left by standard greedy dispatchers. By achieving a double-digit efficiency gain in a high-variance environment, the algorithm proves to be a commercially viable engine for real-mileage reduction in dynamic, value-sensitive delivery fleets.

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| Data Availability Statement

The production-ready JavaScript engine and implementation source code for the IVCP Route Optimizer are hosted exclusively at the official project repository:
https://github.com/AEjonanonymous/IVCP-Route-Optimizer
The repository is published under the copyleft **GNU Affero General Public License (AGPL-3.0)** for non-commercial research, academic evaluation, and open-source verification. Any commercial utilization, proprietary integration, or closed-source deployment requires a separate commercial license from the author.

| References

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